

TuitionBureau

Complete Project & Business Workflow Documentation

Enterprise Tuition Bureau Management, Automation & AI Operations Platform

1. Project Overview

TuitionBureau is an enterprise tuition bureau management and automation platform designed to manage the complete lifecycle of tuition services—from the first parent enquiry through CRM, tuition requirements, tutor matching, assignment, scheduling, sessions, attendance, billing, payments, tutor payables, communication, reporting, automation and AI-powered operations.

2. Main Objective

The goal is to replace disconnected spreadsheets, manual follow-ups, scattered communication and manual reporting with one centralized system. The platform acts as an operating system for a tuition bureau.

3. Users & Roles

Internal users: SUPER_ADMIN, ORG_ADMIN, BRANCH_MANAGER, COORDINATOR, FINANCE_MANAGER and SUPPORT_AGENT. External users: TUTOR, PARENT and STUDENT.

4. CRM & Lead Workflow

A potential customer contacts the tuition bureau. A lead is created, qualified, followed up and converted into a tuition request when the requirement is confirmed.

5. Tuition Request

A tuition request represents the actual service requirement, such as a Class 10 Mathematics home-tuition requirement with a location, schedule and budget.

6. Tutor Matching

The system finds eligible tutors using hard filters and deterministic scoring. Subject, grade, teaching mode, experience, rate, location, availability, preferences and profile quality can influence ranking. AI must not override hard eligibility rules.

7. Tutor Assignment

Staff reviews recommended tutors and selects a candidate. The system then creates the official assignment containing student, parent, tutor, subject, grade, mode, dates, session frequency, duration, rates and location.

8. Scheduling & Sessions

Assignments are converted into scheduled sessions. The system manages dates, times, recurring sessions, cancellations and rescheduling.

9. Attendance & Session Notes

Tutors record attendance for authorized sessions. Typical states are PRESENT, LATE, ABSENT and EXCUSED. Tutors can also record appropriate session/learning notes. Attendance becomes an important input to billing and tutor payable calculations.

10. Billing & Payments

Billable sessions flow into billing periods and invoices. The system supports invoice line items, partial payments, payment allocation, outstanding balances and overdue invoices. Financial values use precise decimal types.

11. Tutor Payables

Tutor compensation is calculated separately from parent billing. Payables can be approved and then paid, allowing the bureau to track revenue, tutor costs and business margin.

12. Communication

Internal notifications and messaging support daily operations. Phase 8 provides provider-agnostic Email, SMS and WhatsApp communication with consent, preferences, delivery tracking, retries, webhooks and outbox processing.

13. Reporting & Analytics

Phase 6 provides reporting for CRM, tuition requests, tutor matching, assignments, students, tutors, scheduling, attendance, finance, tutor payables, operations, branches, staff and communication.

14. Parent Portal

Parents can access their own dashboard, children, tuition requests, assignments, tutors, schedule, sessions, attendance, invoices, payments, messages, notifications, profile and settings.

15. Tutor Portal

Tutors can access their profile, students connected through assignments, assignments, schedule, sessions, attendance, notes, earnings/payables, payments, messages, notifications, availability and settings. Internal staff-only information remains protected.

16. Student Portal

Students can access their dashboard, profile, tutor, assignments, schedule, sessions, attendance, learning overview, messages, notifications and settings.

17. Staff / Operations Portal

Staff coordinate leads, tuition requests, tutor matching, assignments, scheduling, attendance follow-ups, customer communication, operational tasks, escalations and finance follow-ups according to permissions.

18. Phase 7 — Workflow Automation

Phase 7 provides deterministic automation for lead follow-ups, tuition-request reminders, assignment follow-ups, session reminders, attendance follow-ups, overdue invoice workflows, tutor-payable workflows, staff tasks and escalation rules.

19. Phase 8 — Omnichannel Communication

Phase 8 connects workflows and business services to external communication providers. The business layer remains provider-agnostic and communication status is tracked through queues/outbox and webhook processing.

20. Phase 9 — AI-Assisted Automation

Phase 9 introduces controlled AI assistance for lead summaries, qualification, follow-up drafts, tuition-request extraction, tutor-match explanations, assignment risk analysis, attendance analysis, communication drafting, support classification, finance explanations and reporting insights.

21. Phase 10 — Autonomous AI Operations

Phase 10 introduces controlled autonomous agents. Agents can monitor business events, reason over authorized context, plan bounded actions, execute approved tools and escalate when human intervention is required.

22. Phase 10 Agents

Core agents include Lead Operations, Tutor Operations, Scheduling, Attendance, Finance Follow-up, Support, Communication Monitoring, Daily Operations and Management Intelligence.

23. Agent Safety

The AI agent is not the authority. The existing platform remains the source of truth. Every action follows authorization, tenant/branch checks, business rules, agent policy, risk checks, limits and approved tools.

24. Agent Control Center

Administrators can view active agents, running/completed runs, pending approvals, actions, errors, AI usage, estimated cost, policies, schedules, rate limits and emergency controls.

25. Human Approval

Medium- and high-risk actions can require human approval. Staff see what the agent wants to do, why, which record is affected, risk level and proposed action before approving.

26. Guardrails

Agents use maximum steps, runtime, tool calls, action limits, communication limits, AI budgets, cooldowns, concurrency controls, idempotency, loop detection, circuit breakers and kill switches.

27. Tenant & Branch Security

Every agent run is tenant-aware and branch-aware. Organization and branch scope comes from trusted server-side context. Cross-tenant and unauthorized cross-branch access must be denied and audited.

28. Prompt Injection Protection

Emails, SMS, WhatsApp messages, lead notes, tutor bios and other user-generated content are untrusted. Embedded instructions must never change agent permissions, tools, tenant scope or security policy.

29. Finance Safety

Finance agents can monitor invoices, balances and overdue status and can generate permitted reminders. They must not independently refund payments, create payments, modify invoice totals, change payables or alter authoritative financial records.

30. Existing Phase Integration

Phase 5 provides tasks/escalations; Phase 6 provides reporting; Phase 7 provides triggers/schedules/workflows; Phase 8 provides external communication; Phase 9 provides AI providers, context, tools, prompts, action proposals and usage tracking.

31. Complete Customer Lifecycle

The complete lifecycle is: Parent/Customer → Lead → Qualification → Tuition Request → Tutor Matching → Staff Selection → Assignment → Scheduling → Sessions → Attendance → Session Notes → Billing → Invoice → Payment → Tutor Payable → Tutor Payment → Reporting → Automation → AI Assistance → Autonomous AI Operations.

32. Technical Architecture

The platform uses React + TypeScript + Vite, Tailwind/shadcn UI, TanStack Query, Node.js + Express + TypeScript, business services, Prisma and PostgreSQL, supported by authentication, RBAC, tenant isolation, audit logs, workflows, communications, AI, agents and reporting.

33. Development & Security Principles

Do not duplicate existing systems. Never trust client tenant IDs or resource IDs. Do not expose secrets. Do not allow arbitrary SQL, code execution, unrestricted HTTP/browser access or unrestricted filesystem access. Use transactions, indexes, DTOs, validation and audit logging.

34. Testing

Test agent registry, policy engine, planner, tool executor, run lifecycle, rate limits, budgets, approvals, kill switch, tenant isolation, branch isolation, IDOR, prompt injection, duplicate communication, concurrency, loop protection and financial restrictions.

35. Recommended Rollout

Start with agents OFF, then Lead Agent DRY RUN, then SUPERVISED mode, then limited low-risk autonomy. Add communication, attendance, tutor operations, finance follow-up and scheduling agents gradually based on measured reliability.

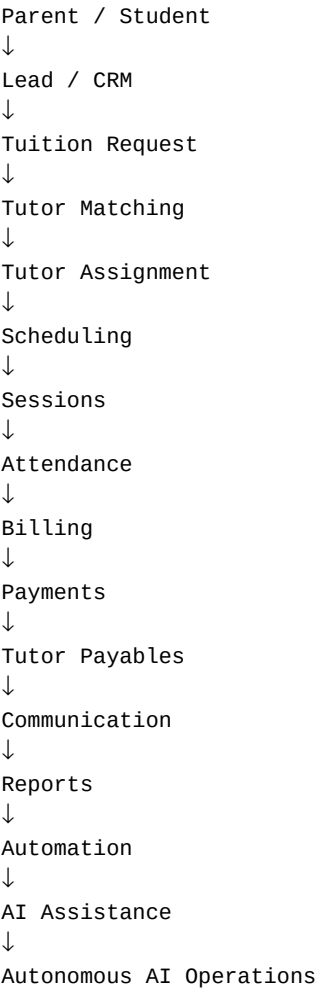
36. Project Philosophy

TuitionBureau follows a layered architecture: deterministic business rules first, automation second, communication infrastructure third, AI assistance fourth and controlled autonomous agents last. AI increases operational efficiency without becoming the security or business-rule authority.

37. One-Line Explanation

TuitionBureau is an end-to-end tuition bureau operating system that manages leads, tuition requirements, tutor matching, assignments, scheduling, sessions, attendance, billing, payments, communication, reporting, workflow automation and AI-powered operations with enterprise security and controlled autonomous agents.

38. Core Workflow Diagram



39. Layered Architecture

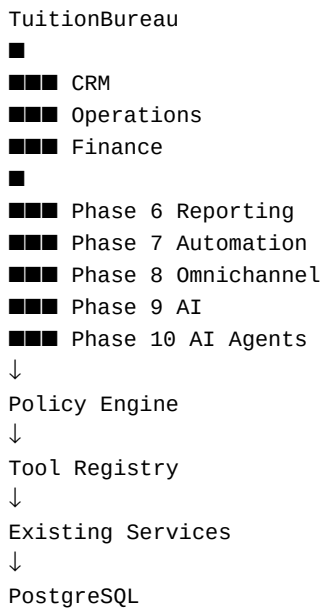
- Layer 1 – Core SaaS
Users • Organizations • Branches • Roles • Permissions
- Layer 2 – Tuition Operations
Leads • Requests • Tutors • Students • Assignments • Scheduling • Sessions • Attendance
- Layer 3 – Commercial Operations
Invoices • Payments • Tutor Payables • Commissions • Reports
- Layer 4 – Automation
Workflows • Tasks • Notifications • Email • SMS • WhatsApp
- Layer 5 – Intelligence
AI Assistance • Recommendations • AI Agents • Autonomous Operations

40. Phase Roadmap

Phase	Purpose
1	Foundation, authentication, RBAC and multi-tenancy
2A–2F	People, tuition requests, CRM, tutor matching and assignments
3A–3D	Scheduling, attendance, billing, notifications and messaging
4A–4C	Parent, Tutor and Student portals
5	Staff and operations
6	Advanced reporting and analytics

Phase	Purpose
7	Workflow automation
8	Email, SMS and WhatsApp omnichannel communication
9	AI-assisted automation
10	Controlled autonomous AI agents and intelligent operations

41. Final Architecture



Document purpose: Project handover, business workflow reference, architecture overview and stakeholder documentation.